

L04 Linear Programming

CS 169 / CS 268
Introduction to Optimization

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A minimum with a nonzero derivative

Constraints can determine where an optimum occurs.

$$\text{minimize}_{x \geq 0} f(x) = x.$$

$$x^* = 0, \quad f(x^*) = 0, \quad f'(x^*) = 1.$$

At the boundary, a decreasing direction may leave the feasible set.

What is a linear program?

A **linear program (LP)** optimizes a linear objective under linear equalities and inequalities.

$$\begin{array}{ll}\text{maximize} & c^T x \\ \text{subject to} & Ax \leq b, \quad x \geq 0.\end{array}$$

$x \in \mathbb{R}^n$ is the decision vector. The coefficients A , b , and c are given.

Variables are continuous. A constant added to the objective does not change the optimizers.

A production-planning problem

A producer makes two divisible products. Let x_1, x_2 be their amounts, measured in units.

Product 1 earns 3 per unit. Product 2 earns 2 per unit.

Each unit uses one machine-hour. There are 4 machine-hours available.

At most 2 units of product 1 can be sold. Both production amounts must be nonnegative.

The production linear program

$$\begin{array}{ll}\text{maximize} & 3x_1 + 2x_2 \\ \text{subject to} & x_1 + x_2 \leq 4, \\ & x_1 \leq 2, \\ & x_1, x_2 \geq 0.\end{array}$$

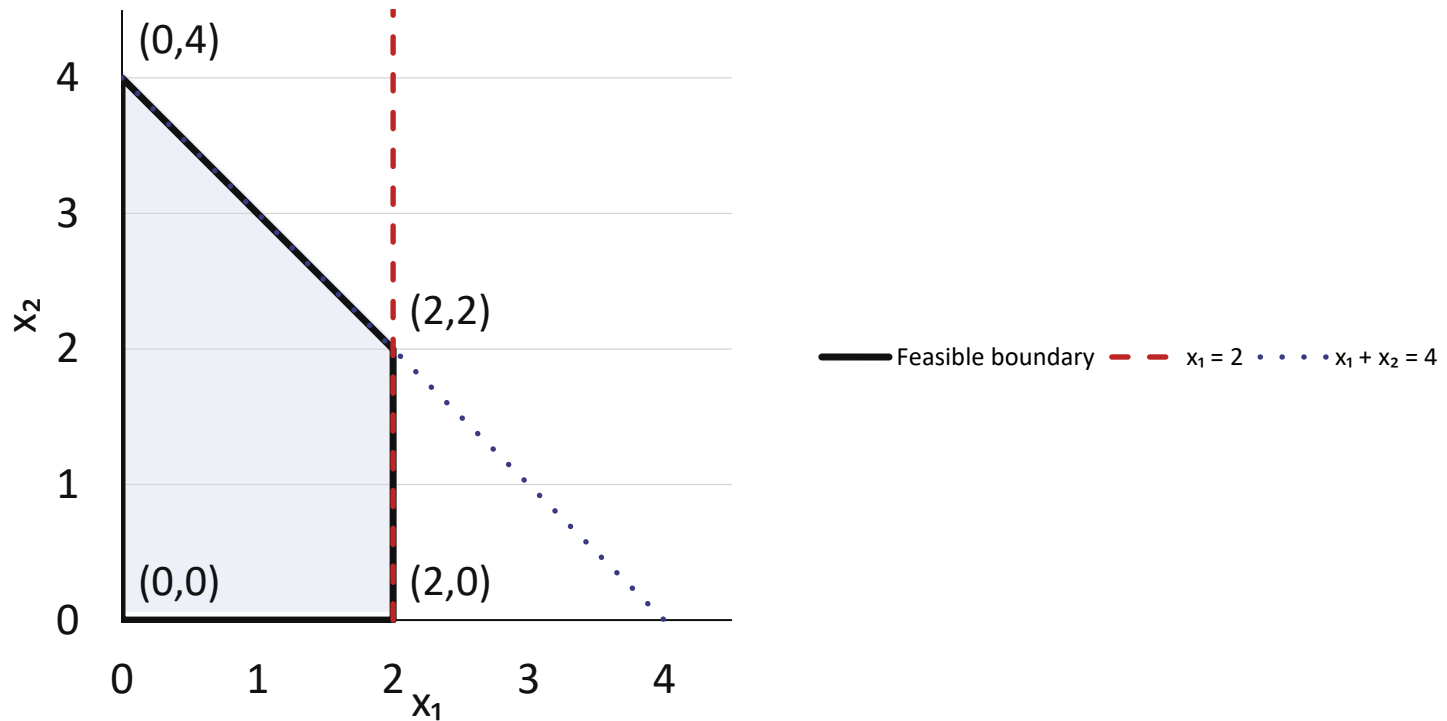
A point is **feasible** if it satisfies every constraint.

$$(1, 1) \text{ is feasible, } \quad 3(1) + 2(1) = 5.$$

$(3, 0)$ violates the product-1 limit, even though it uses only 3 machine-hours.

The feasible region

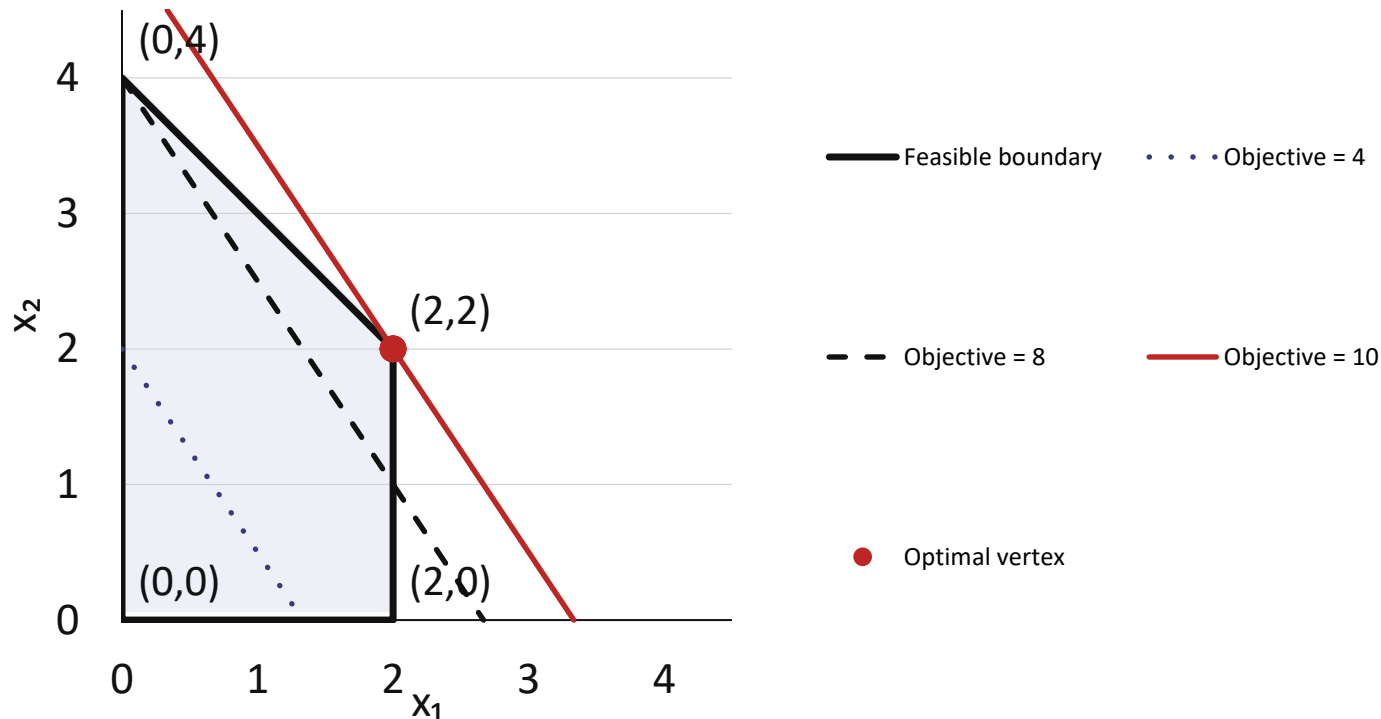
Each inequality selects a half-plane. Their intersection is the **feasible region**.



All points on or inside the solid polygon are feasible.

Objective level lines

$$3x_1 + 2x_2 = k, \quad k = 4, 8, 10.$$



Higher k means higher profit. The last feasible contact is $(2,2)$.

A proof of optimality

Every feasible point satisfies $x_1 + x_2 \leq 4$ and $x_1 \leq 2$.

$$3x_1 + 2x_2 = 2(x_1 + x_2) + x_1 \leq 2 \cdot 4 + 2 = 10.$$

$$x^* = (2, 2), \quad 3x_1^* + 2x_2^* = 10.$$

An active constraint holds with equality. Here both resource constraints are active at the optimum.

Why the feasible region is convex

A **polyhedron** is an intersection of finitely many closed half-spaces. A bounded polyhedron is a **polytope**.

If $Ax \leq b$ and $Ay \leq b$, then for $0 \leq \lambda \leq 1$,

$$A(\lambda x + (1 - \lambda)y) = \lambda Ax + (1 - \lambda)Ay \leq b.$$

Nonnegativity is preserved too. Thus the entire segment between feasible points remains feasible.

Vertices and extreme points

Let P be the feasible region. A point $v \in P$ is a **vertex** if it cannot lie strictly between two distinct feasible points.

$$v = \lambda u + (1 - \lambda)w, \quad 0 < \lambda < 1, \quad u, w \in P$$

For a vertex, this forces $u = w = v$. Vertices are also called **extreme points**.

$$(1, 1) = \frac{1}{2}(0, 0) + \frac{1}{2}(2, 2) \quad \text{is not a vertex.}$$

An optimum at a vertex

Theorem. Let P be a nonempty bounded polyhedron. A linear objective attains a maximum at at least one vertex of P .

$$\max_{x \in P} c^T x = \max_{v \text{ a vertex of } P} c^T v.$$

The same statement holds for minimization. It does not require the optimizer to be unique.

Why vertices suffice

Every point of a nonempty polytope is a convex combination of its vertices $v_1, \dots, v_N$.

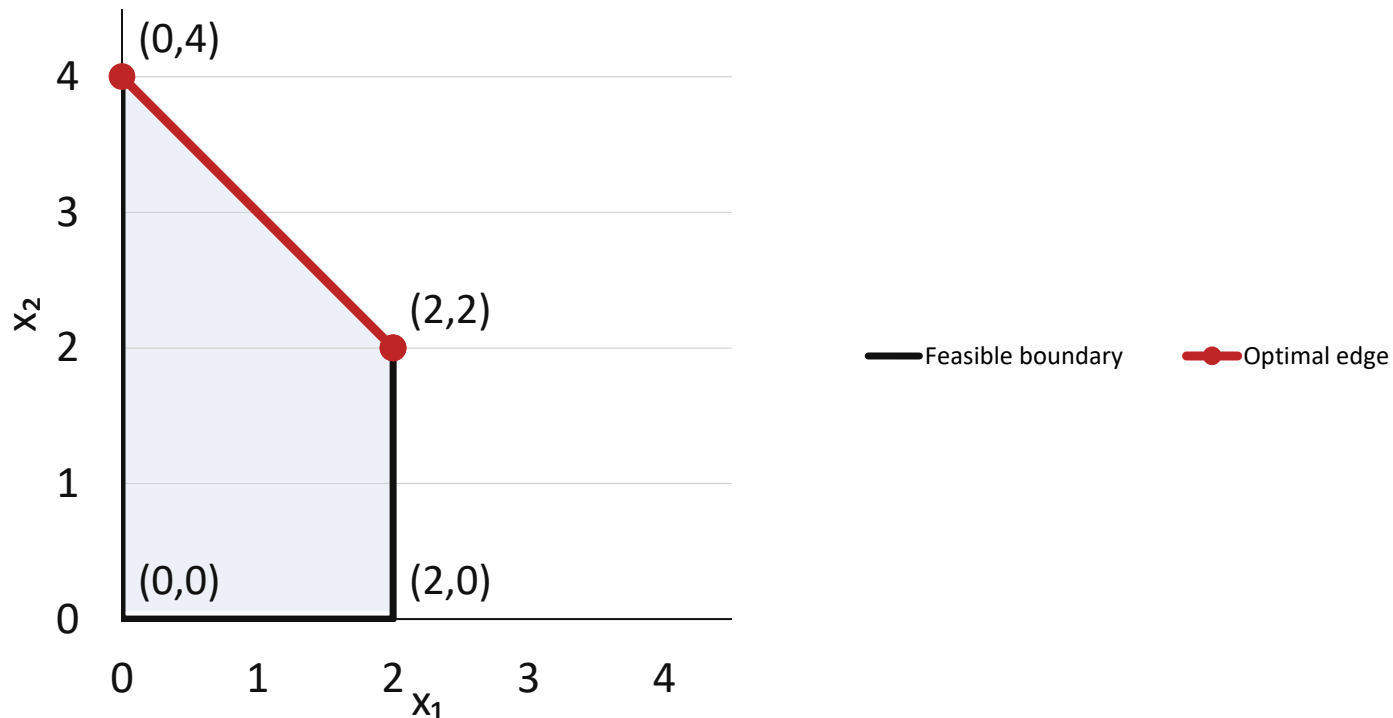
$$x = \sum_{i=1}^N \lambda_i v_i, \quad \lambda_i \geq 0, \quad \sum_{i=1}^N \lambda_i = 1.$$

$$c^\top x = \sum_{i=1}^N \lambda_i c^\top v_i \leq \max_i c^\top v_i.$$

A weighted average cannot exceed its largest entry.

An entire edge can be optimal

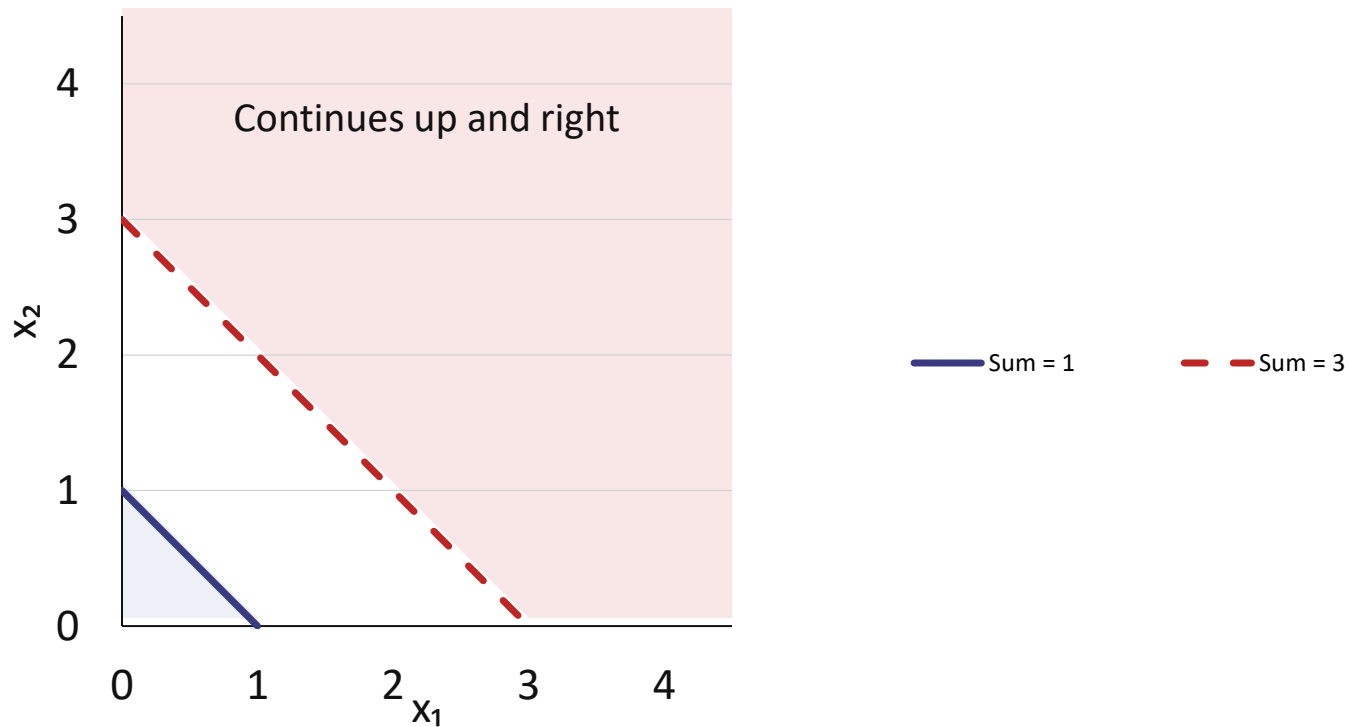
Keep the constraints, but maximize $x_1 + x_2$ instead.



$$x = (t, 4 - t), \quad 0 \leq t \leq 2 : \quad x_1 + x_2 = 4.$$

Infeasibility

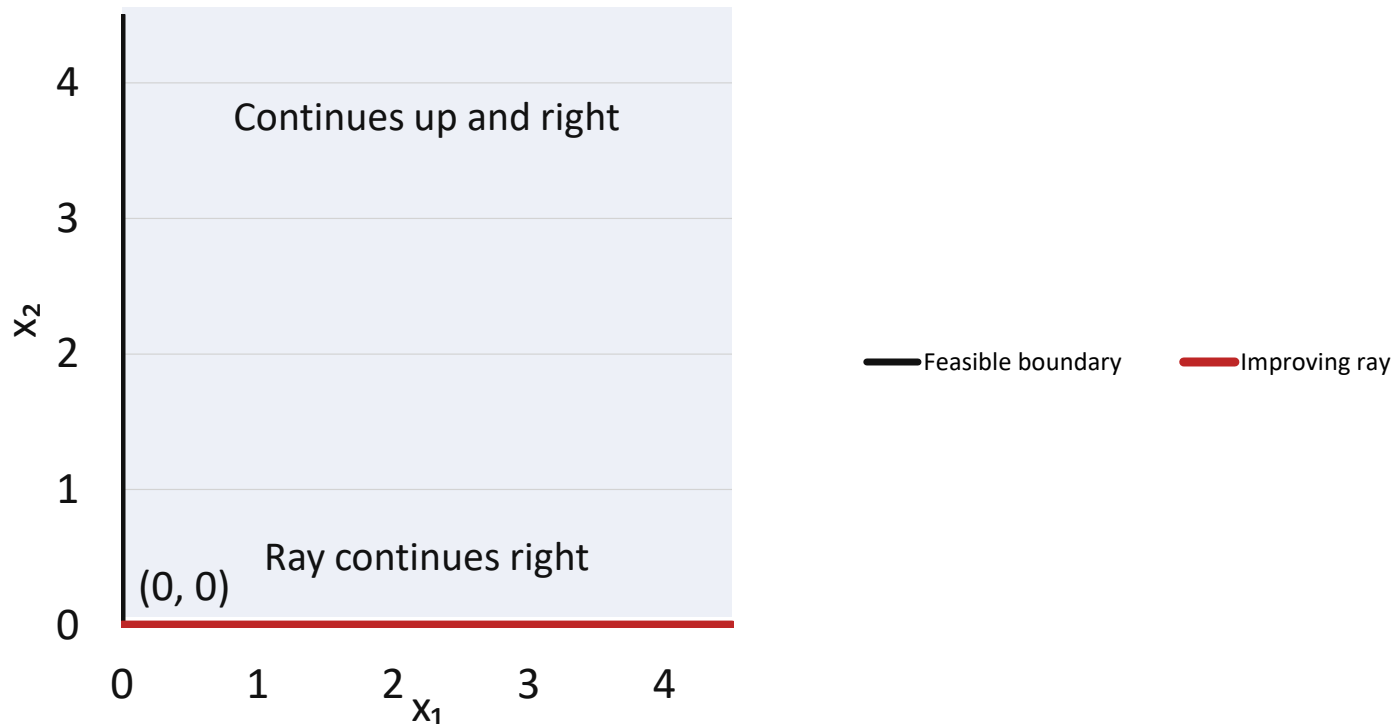
$$x_1 + x_2 \leq 1, \quad x_1 + x_2 \geq 3, \quad x_1, x_2 \geq 0.$$



The two regions do not overlap. No point satisfies all constraints.

An unbounded objective

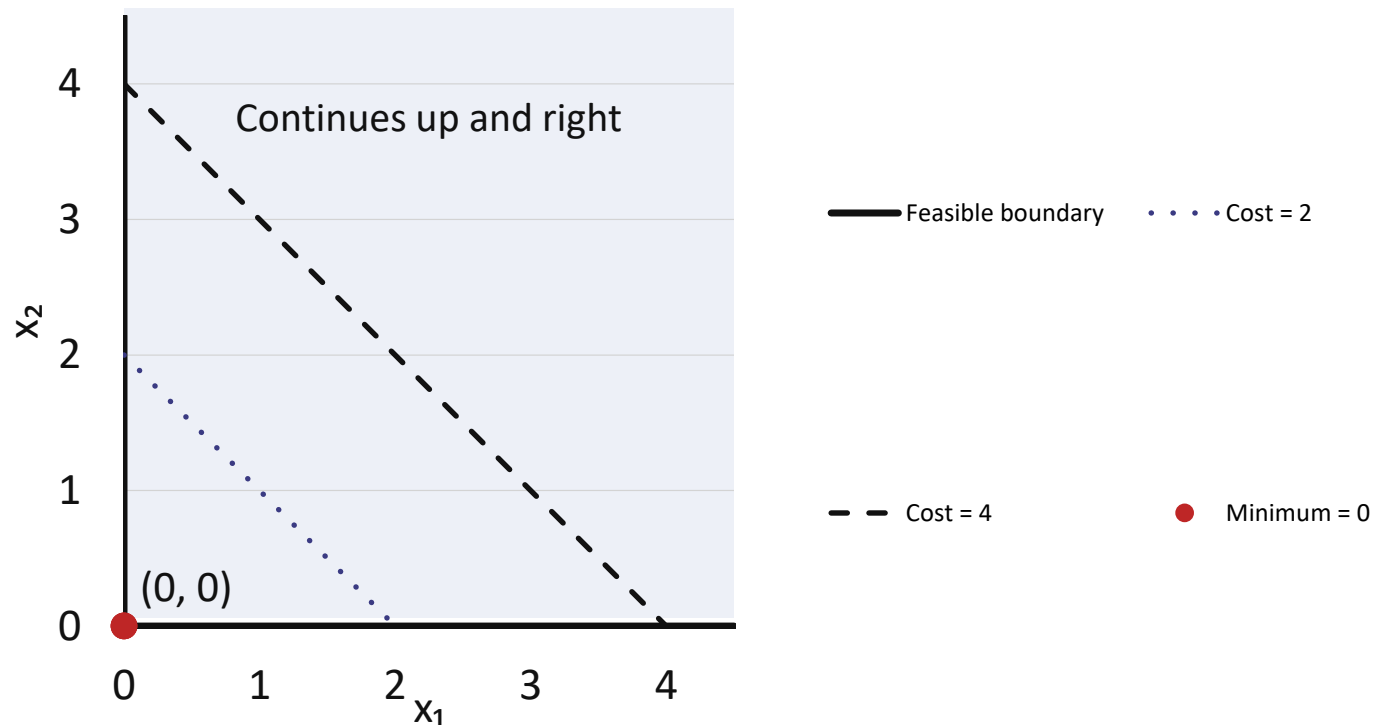
$$\max\{3x_1 + 2x_2 : x_1, x_2 \geq 0\}.$$



$$x = (t, 0), \quad t \geq 0 : \quad 3x_1 + 2x_2 = 3t \longrightarrow +\infty.$$

An unbounded region can have an optimum

$$\min\{x_1 + x_2 : x_1, x_2 \geq 0\}.$$



$$x_1 + x_2 \geq 0, \quad x^* = (0, 0), \quad f(x^*) = 0.$$

A minimum-cost nutrition model

Choose amounts $u, v \geq 0$ of two divisible foods, costing 2 and 3 per portion.

Nutrient A: $2u + v$, with at least 5 required.

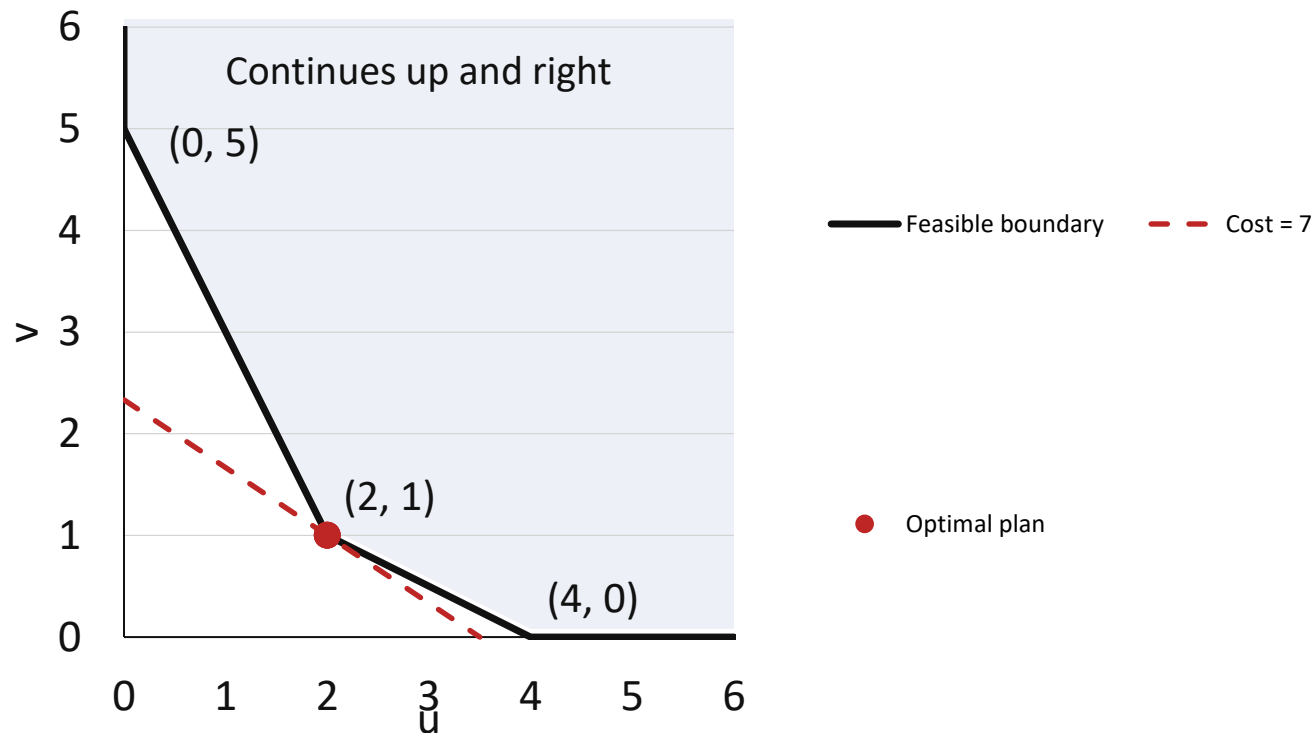
Nutrient B: $u + 2v$, with at least 4 required.

$$\begin{array}{ll}\text{minimize} & 2u + 3v \\ \text{subject to} & 2u + v \geq 5, \\ & u + 2v \geq 4, \\ & u, v \geq 0.\end{array}$$

Minimum requirements produce $\geq$ constraints.

The minimum-cost vertex

$$\min(2u + 3v), \quad 2u + v \geq 5, \quad u + 2v \geq 4, \quad u, v \geq 0.$$



$$(u^*, v^*) = (2, 1), \quad 2u^* + 3v^* = 7.$$

What makes the formulation linear?

Coefficients are fixed data. Decision variables appear only to the first power and do not multiply each other.

$2u + v \geq 5$ is linear, $uv \geq 5$ is not.

Additivity and constant per-unit contributions are modeling assumptions.

Requiring whole numbers changes an LP into an integer program. We will study that next week.

Standard form

We will use the following **standard form**:

$$\begin{array}{ll}\text{minimize} & c^T z \\ \text{subject to} & Az = b, \\ & z \geq 0.\end{array}$$

All constraints are equalities except nonnegativity. Extra variables can encode inequalities.

Maximizing $p(x)$ is equivalent to minimizing $-p(x)$.

Slack variables

A **slack variable** measures unused capacity.

$$x_1 + x_2 \leq 4 \quad \Longleftrightarrow \quad x_1 + x_2 + s_1 = 4, \quad s_1 \geq 0.$$

$$x_1 \leq 2 \quad \Longleftrightarrow \quad x_1 + s_2 = 2, \quad s_2 \geq 0.$$

A constraint is active exactly when its slack is zero. At $(1, 1)$, $s_1 = 2$ and $s_2 = 1$.

The production LP in standard form

$$z = \begin{pmatrix} x_1 \\ x_2 \\ s_1 \\ s_2 \end{pmatrix}, \quad c = \begin{pmatrix} -3 \\ -2 \\ 0 \\ 0 \end{pmatrix}.$$

$$\underbrace{\begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}}_A z = \underbrace{\begin{pmatrix} 4 \\ 2 \end{pmatrix}}_b, \quad z \geq 0.$$

$$\min c^T z = \min(-3x_1 - 2x_2).$$

$z^* = (2, 2, 0, 0)^T$ has minimum cost -10 and profit 10 .

Other standard-form conversions

A lower bound: subtract a nonnegative surplus.

$$2u + v \geq 5 \quad \Longleftrightarrow \quad 2u + v - s = 5, \quad s \geq 0.$$

A variable unrestricted in sign: use a difference.

$$t = t^+ - t^-, \quad t^+, t^- \geq 0.$$

Substitute this difference everywhere, including the objective.

Basic feasible solutions

For $Az = b$, assume A has m rows, all linearly independent. Choose m independent columns, indexed by B .

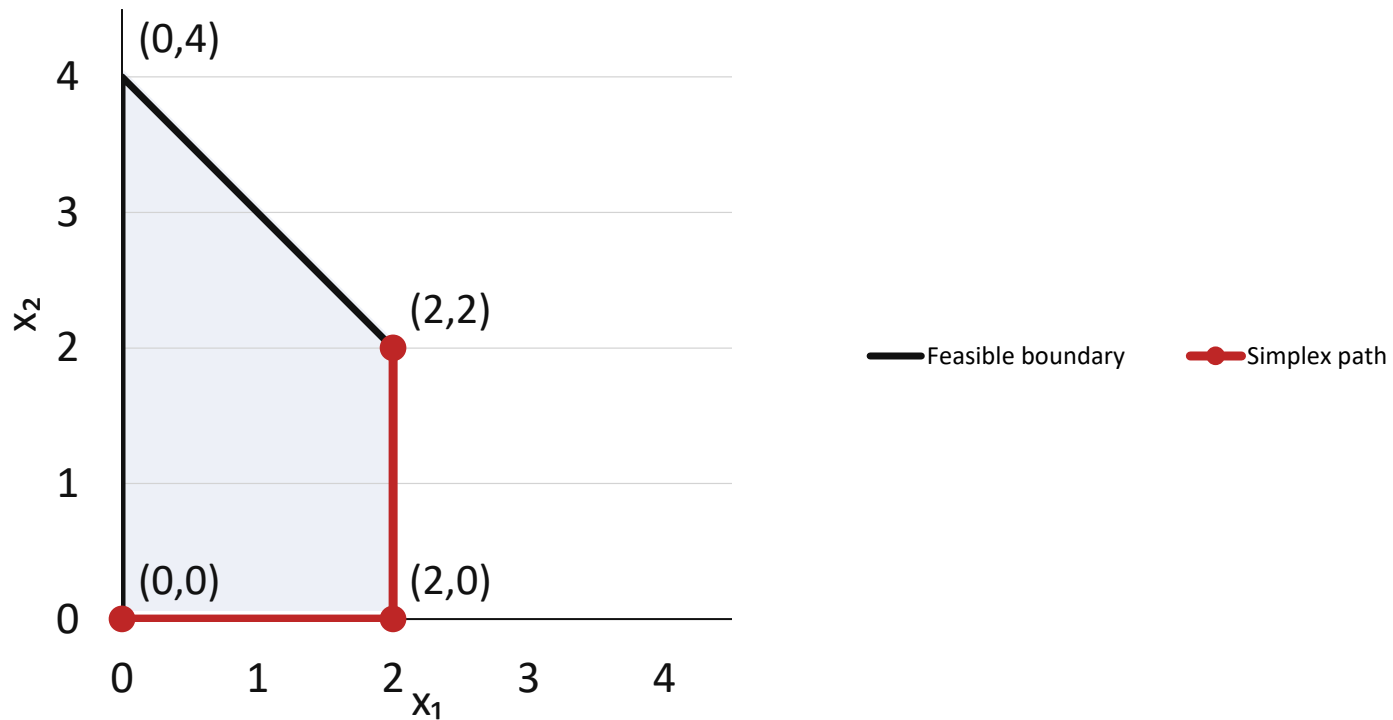
$$z_j = 0 \ (j \notin B), \quad A_B z_B = b.$$

If the resulting $z_B \geq 0$, this is a **basic feasible solution**. These are the vertices in standard form.

$$x_1 = x_2 = 0 \quad \Longrightarrow \quad s_1 = 4, \quad s_2 = 2.$$

A geometric preview of simplex

$(0,0) \longrightarrow (2,0) \longrightarrow (2,2)$, profit: 0, 6, 10.



Simplex updates a basis while maintaining feasibility.

Reading and next lecture

Reading: Sections 15.2–15.5 for LP models, geometry, and standard form.

Further reading: Sections 15.6–15.8 for basic solutions and their geometry.

Next lecture: the simplex method and an introduction to LP duality (Chapters 16–17).

Questions to check understanding

1. If two different feasible points are optimal for an LP, what can you say about every point on the segment between them? Why?
2. Does an unbounded feasible region force the objective to be unbounded? What additional information matters?